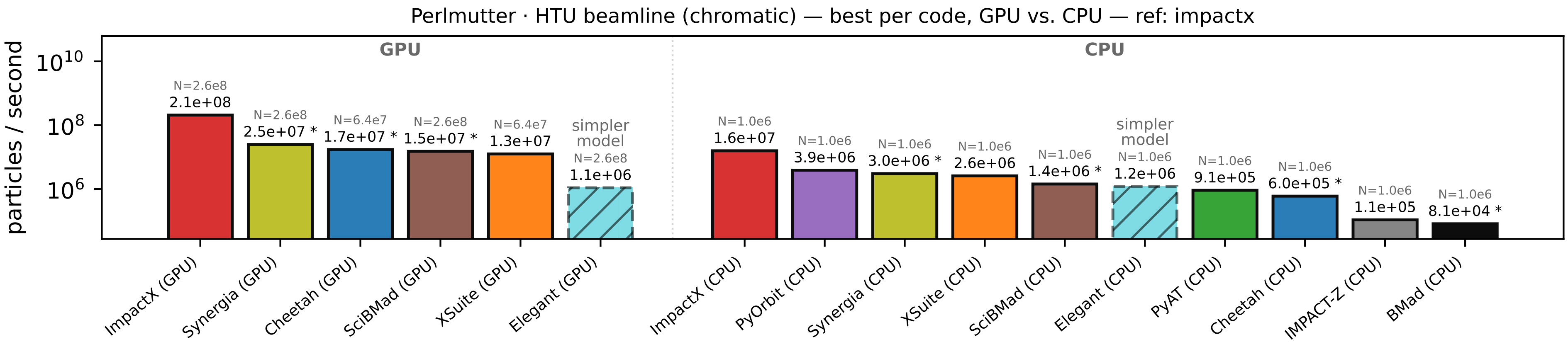

'simpler model' = a lower-fidelity model vs the reference — elegant: tracks geometric slope (x,x') not canonical (x,px)

** Cheetah/SciBmad/Bmad/Synergia have no chromatic-paraxial model here; they run the costlier exact map (agrees within tol)*

each bar = the fastest measured config for that code on that device (any precision / fast-math); N = the particle count that bar was measured at (top of each device's sweep)

hardware: CPU: AMD EPYC 7763 (64 cores) GPU: NVIDIA A100-SXM4-40GB

versions: impactx 26.08 · synergia v2024.04-209-g795b99f39 · cheetah 0.8.2 · scibmad ? · xsuite 0.53.4 · elegant elegant-2026.4.0 · pyorbit 96c8def · pyat 0.8.0 · impactz V2.7.7 · bmad 20260904-1